

MERN STACK

DAY – 20

17 July 2026

React

Fetch API

The Fetch API is a built-in JavaScript method used to request data from a server or API. It is commonly used to perform GET, POST, PUT, and DELETE operations.

Features

- Built into JavaScript (no extra library needed).
- Returns a Promise.
- Supports asynchronous programming with `.then()` or `async/await`.
- Used to communicate with REST APIs.

Fetch API using async/await

`async` and `await` are JavaScript keywords used to work with asynchronous operations (such as fetching data from an API). They make asynchronous code easier to read and write.

- `async`: Declares a function as asynchronous. An `async` function always returns a Promise.
- `await`: Waits for a Promise to resolve before continuing to the next line of code. It can only be used inside an `async` function.

Syntax

```
async function functionName() {  
  try {  
    const response = await fetch("API_URL");  
    const data = await response.json();  
    console.log(data);  
  } catch (error) {  
    console.log(error);  
  }  
}
```

```
functionName();
```

TASK-1

```
import { useEffect, useState } from "react";

export const Fetti = () => {
  const [val, setval] = useState([]);
  useEffect(() => {
    getdata();
  }, [])
  const getdata = async () => {
    const res = await fetch("https://jsonplaceholder.typicode.com/posts");
    if (res.ok) {
      const v = await res.json();
      if (v) {
        alert("DATA FETCHED .....");
        setval(v);
      }
    } else {
      alert("FAILURE IN DATA FETCHING");
    }
  }
  return (
    <>
    <div className="container mt-3">
    <table className="table table-bordered table-hover">
    <thead className="table-primary">
    <tr>
    <th>USER ID</th>
    <th>ID</th>
    <th>TITLE</th>
    <th>BODY</th>
    </tr>
    </thead>

    <tbody>
    {val.map((a, index) => (
      <tr key={index}>
      <td>{a.userId}</td>
      <td>{a.id}</td>
      <td>{a.title}</td>
      <td>{a.body}</td>
      </tr>
    ))}
    </tbody>
    </table>
    </div>
    </>
  );
}
```

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TASK-2 BILL GENERATOR

```
import { useState } from "react";

export const Gst = () => {
  const [name, setName] = useState("");
  const [billName, setBillName] = useState("");

  const [act, setact] = useState("");
  const [discount, setdis] = useState("");
  const [dp, setsp] = useState("");
  const [gst, setgst] = useState("");
  const [f, setf] = useState("");
  const [acc, setacc] = useState("");

  const gen = () => {
    const p = Number(act);
    const ac = Number(acc);
    setBillName(name);

    const t = p + ac;

    if (t < 40000) {
      setdis(0);
      setsp(t);
      const g = t * 0.18;
      setgst(g);
      setf(t + g);
    } else if (t >= 40000 && t < 50000) {
      setdis(10);
      const price = t - (t * 10) / 100;
      setsp(price);
      const g = price * 0.18;
      setgst(g);
      setf(price + g);
    } else {
      setdis(15);
      const price = t - (t * 15) / 100;
      setsp(price);
      const g = price * 0.18;
      setgst(g);
      setf(price + g);
    }
  };

  return (
    <div className="container mt-5">
      <div className="card shadow p-4">
        <h2 className="text-center text-primary mb-4">
          GST BILL GENERATOR
        </h2>

        <div className="mb-3">
          <label className="form-label">NAME</label>
          <input
            className="form-control"
            type="text"

```

```
            type="text"
            onChange={(e) => setName(e.target.value)}
          </input>
        </div>

        <div className="mb-3">
          <label className="form-label">LAPTOP MRP</label>
          <input
            className="form-control"
            type="number"
            onChange={(e) => setact(e.target.value)}
          </input>
        </div>

        <div className="mb-3">
          <label className="form-label">ACCESSORIES MRP</label>
          <input
            className="form-control"
            type="number"
            onChange={(e) => setacc(e.target.value)}
          </input>
        </div>

        <button className="btn btn-success w-100" onClick={gen}>
          GENERATE BILL
        </button>

        <hr />

        <div className="card bg-light p-3 mt-3">
          <h4 className="text-center text-success">Bill Details</h4>

          <p><b>Name:</b></b> {billName}</p>
          <p><b>Discount:</b></b> {discount}%</p>
          <p><b>Selling Price:</b></b> ₹{dp}</p>
          <p><b>GST (18%):</b></b> ₹{gst}</p>

          <h5 className="text-danger">
            Final Amount: ₹{f}
          </h5>
        </div>
      </div>
    </div>
  );
};
```

GST BILL GENERATOR

NAME

DELL - G15 GAMING LAPTOP

LAPTOP MRP

59000

ACCESSORIES MRP

20000

GENERATE BILL

Bill Details

Name: DELL - G15 GAMING LAPTOP

Discount: 15%

Selling Price: ₹67150

GST (18%): ₹12087

Final Amount: ₹79237